The Effect of Conservative-Coded TikTok Content on Exposure to Explicit Political Media

POLSGU4724 | Pre-Analysis Plan

Katie Keil

2026-11-07

Recent anecdotal evidence (Liscomb 2025) has alluded to the existence of an alt-right "pipeline" on social media platforms, whereby apolitical social and/or lifestyle content may contain references to ideologically-extreme viewpoints. Though previous research has studied this phenomenon in the context of older platforms like YouTube (Brown et al. n.d.) and Twitter (now X; Huszár et al. (2022)), this project examines the alt-right pipeline as seen on TikTok, a social media platform that prioritizes short-form video content.

The designed experiment seeks to determine whether exposure to social content that aligns with conservative ideological themes leads to a greater propensity for Tiktok's algorithmic feed design to present users with social and political content that directly implicates conservative ideological beliefs. A secondary area of inquiry is whether variations in the number of TikTok videos containing conservative-coded social content alter the algorithm's presentation frequency of explicitly, politically conservative content. Therefore, I propose the following two hypotheses:

> **Hypothesis 1**: TikTok user accounts who interact with conservative-coded social content will have a greater frequency of explicitly politically-conservative content on their "For You" page than TikTok user accounts who interact with apolitical short-form content.

> **Hypothesis 2**: TikTok user accounts who interact with more conservative-coded social content overall will have a greater frequency of explicitly politically-conservative content on their "For You" page than TikTok user accounts who interact with less conservative-coded social content overall.

Empirical Strategy

Subjects

For this experiment, I will create 30 independent TikTok accounts on [DATE], each of which represent one individual user profile. All accounts will have zero followers and follow zero other accounts to minimize the potential confounding role of a user's social network on the types of recommended social content. Furthermore, each account will only interact with other videos by double-tapping, or "liking," that video without leaving any comments. Each username will consist of the same base name, `notoriousdpg`, followed by a random number from 1 to 30.

As TikTok account guidelines state that any email address, phone number, or social media account can only be linked to one username at a time (TikTok support n.d.), I will create 30 randomly-generated, unique email addresses for each account using Apple iCloud's Hide My Email function (Create and manage hide my email addresses in settings on iPhone n.d.). I shall do the same for all account passwords using Apple's Passwords app. The following table outlines the usernames, email addresses, and passwords generated for each TikTok account:

Table 1: Account Information

Username	Email Address	Password
notoriousdpg1		
notoriousdpg2		
notoriousdpg3		
notoriousdpg4		
notoriousdpg5		
notoriousdpg6		
notoriousdpg7		
notoriousdpg8		
notoriousdpg9		
notoriousdpg10		
notoriousdpg11		
notoriousdpg12		
notoriousdpg13		
notoriousdpg14		
notoriousdpg15		
notoriousdpg16		
notoriousdpg17		
notoriousdpg18		
notoriousdpg19		
notoriousdpg20		
notoriousdpg21		

Username	Email Address	Password
notoriousdpg22		
notoriousdpg23		
notoriousdpg24		
notoriousdpg25		
notoriousdpg26		
notoriousdpg27		
notoriousdpg28		
notoriousdpg29		
notoriousdpg30		

Treatment

I define my treatment as a set of three hashtags, each of which corresponds to a given conservative-coded social topic as defined by Liscomb (2025). The first hashtag, #cleanbeauty, intends to measure whether user accounts seeking content on healthy living will be recommended adjacent content about the Make America Healthy Again (MAHA) movement, which is a cornerstone of the second Trump administration. The second hashtag, #tradwife, intends to measure whether user accounts seeking content about parenting and motherhood will be recommended adjacent content about traditional gender roles, anti-feminism, and religious ideologies—the latter of which corresponds to the prevalence of Mormon women posting under this hashtag. The final hashtag, #softliving, intends to measure whether user accounts seeking content about women balancing work life and home life will be recommended adjacent content about hierarchical gender norms and women as subservient to their husbands.

Meanwhile, for the control condition, I select three hashtags related to cockapoos, a dog breed that crosses a poodle with a cocker spaniel. I define this topic as a placebo control, since I expect that pet-related content should have no discernible algorithmic pathway to political content. Furthermore, each of the three hashtags corresponds to the set of those most frequently used in the descriptions of multiple short-form content videos about cockapoos, of which the sample of video sources originates from those my mother has sent to my family group chat. The below table outlines each of the indicated search term hashtags:

Table 2: Search Term Hashtags by Condition Assignment

Apolitical Topics (Control)	Conservative-Coded Social Topics (Treatment)
#cockapoo	#cleanbeauty
#cockapoosoftiktok	#tradwife
#cockapoolife	#softliving

For each account, regardless of assignment, I will enter the first hashtag into the search

bar on TikTok and double-tap ("like") *exactly* 25 videos. I will do the same for all hashtags listed in the above table based on the account's randomly-assigned condition.

Randomization

I will employ two complete random assignment procedures over the course of this experiment. Under the first randomization, 15 accounts will be assigned to view the apolitical cockapoo content, and 15 accounts will be assigned to view the conservative-coded social content. Under the second randomization, which will be independent of the previous random assignment procedure, 15 accounts will be assigned to view the apolitical cockapoo content and 15 accounts will be assigned to view the conservative-coded social content. The randomization process will not involve blocking, clustering, or multiple treatment arms.

The randomization procedure will be automated using the `randomizr` package on R (Blair et al. 2019), specifically with the function `complete_ra` . As a randomization check, to test the null hypothesis that the randomization procedures are independent of each other, I will conduct randomization inference by regressing the second random assignment on the first random assignment. I will then calculate the resulting F-statistic.

Outcomes

I define my outcome variable as the number of recommended TikToks containing explicitly political content, measured as a function of the first 50 recommended videos on a user account's "For You" page. I will quantify this as a continuous variable, with possible values ranging from 0 to 50.

To ensure measurements are symmetrical between treatment and control, I will measure outcomes only after the treatment assignment and subsequent viewing of 150 videos (25 videos/hashtag $\times 3$ hashtags $\times 2$ random assignments).

Covariates

I will collect data for two covariates, both also representing the number of recommended TikToks containing explicitly political content, measured as a function of the first 50 recommended videos on a user account's "For You" page. One will be measured prior to the first randomization (baseline), and the other will be measured prior to the second randomization. As such, I will conduct two randomization checks by regressing the first and second random assignment on the number of recommended TikToks containing explicitly political content prior to and following the first random assignment, respectively. I will then calculate the resulting F-statistic.

Separately, for each video watched, I will catalog descriptive statistics for the following values:

`likes`: This represents the total number of likes (excluding that of the user account) on that video.

`comments`: This represents the total number of comments (of which the user account will leave none) on that video.

`followers`: This represents the total number of other accounts (excluding that of the user account) that are following the account which posted the video.

To ensure that there are no potential confounding effects of increased time spent on an account's profile page, I will collect these measurements on a separate device. I will then create a composite score for likes, comments, and followers, respectively, by calculating the mean value across all 150 videos. I will apply logarithmic transformation to these variables.

Analysis

The estimand of study is the average treatment effect of viewing conservative-coded social content on the number of "For You" videos containing explicitly political content. It will be estimated using a difference in means estimator.

I will test both hypotheses using a one-tailed test with randomization inference because logic would suggest that any given user account cannot have fewer recommended videos corresponding to explicit political content when compared to a baseline of zero recommended videos. I will then calculate the resulting F-statistic.

I will then adjust for both of my covariates of interest, which represent the number of recommended TikToks containing explicitly political content following each randomization stage. I will use a two-tailed test with randomization inference because I expect there to be some number of recommended videos containing explicitly political content that is greater than zero after any randomization stage. Thus, comparing outcomes between randomization stages will allow me to conclude whether there is an observed "threshold" at which the set of recommended videos is more aligned with a user's indicated preferences. I will then calculate the resulting F-statistic.

R Code

```
#-------------------------------------------
# 1. Load libraries
library(randomizr)
library(estimatr)
library(ggplot2)
library(readxl)
library(dplyr)
library(ri2)
```

```r
# remove all previous objects from the environment
rm(list = ls())

# set random seed
set.seed(080403)

# design constants from the PAP
n_accounts <- 30    # 30 TikTok accounts
n_treat    <- 15    # 15 treatment / 15 control in each randomization
n_fyp      <- 50    # outcome = political videos among first 50 "For You"
 ↪  videos
n_videos   <- 150   # 25 videos/hashtag x 3 hashtags x 2 randomizations
n_sims     <- 1000  # randomization inference simulations

control_tags <- "#cockapoo, #cockapoosoftiktok, #cockapoolife"
treat_tags   <- "#cleanbeauty, #tradwife, #softliving"

# helper: F-statistic from an OLS regression
get_f <- function(formula, data) {
  unname(summary(lm(formula, data = data))$fstatistic["value"])
}
```

```r
#----------------------------------------------
# 2. Conduct first randomization (complete RA, 15/15)
Z1 <- complete_ra(N = n_accounts, m = n_treat)

#----------------------------------------------
# 3. Conduct second randomization (complete RA, 15/15, independent of the
 ↪  first)
Z2 <- complete_ra(N = n_accounts, m = n_treat)

# Merge both randomizations with the account IDs / usernames from Table 1
assignments <- data.frame(
  account  = 1:n_accounts,
  username = paste0("notoriousdpg", 1:n_accounts),
  Z1       = Z1,
  Z2       = Z2
) |>
  mutate(
    # number of rounds of conservative-coded content (0, 1, or 2) -- for
     ↪  H2
    randomize      = Z1 + Z2,
    hashtags1 = if_else(Z1 == 1, treat_tags, control_tags),
```

```
    hashtags2 = if_else(Z2 == 1, treat_tags, control_tags)
  )

# 15/15 in each round, plus the 2x2 cross-tab of the two assignments
# print(table(Z1 = assignments$Z1, Z2 = assignments$Z2))

# save the assignment sheet to follow while running the experiment
write.csv(assignments, "assignments.csv", row.names = FALSE)
```

```
#-------------------------------------------
# 4. Randomization check 1: are the two randomizations independent?

declaration_Z2 <- declare_ra(N = n_accounts, m = n_treat)

balance_fun_1 <- function(data) get_f(Z2 ~ Z1, data)

ri_balance_1 <- conduct_ri(
  test_function = balance_fun_1,
  declaration   = declaration_Z2,
  assignment    = "Z2",
  sims          = n_sims,
  p             = "upper",
  data          = assignments
)
# print(summary(ri_balance_1))
```

```
#-------------------------------------------
# 5. Read the data files and do the data transformations according to the
↪  PAP
#
# Expected structure of the real data:
#   outcomes_data (one row per account):
#     account   -- 1 to 30
#     pol_wave0 -- # explicitly political videos in first 50 FYP videos,
#                  measured right after account creation, BEFORE
#                  randomization 1 (baseline covariate)
#     pol_wave1 -- # explicitly political videos in first 50 FYP videos,
#                  measured after randomization 1 (covariate)
#     pol_wave2 -- # explicitly political videos in first 50 FYP videos,
#                  measured after randomization 2 (outcome)
#   video_data (one row per liked video, 150 per account):
#     account, round (1/2), hashtag, likes, comments, followers

# ----- REAL DATA (uncomment once collected; delete the sample data below)
↪  -----
```

```r
# outcomes_data <- read_excel("practicum_data.xlsx", sheet = "outcomes")
# video_data    <- read_excel("practicum_data.xlsx", sheet = "videos")

# ----- SAMPLE DATA (simulated so the code runs before data collection)
↪  --------
outcomes_data <- assignments |>
  mutate(
    # account-specific baseline propensity to be shown political content
    base_p    = runif(n_accounts, min = 0.02, max = 0.08),
    pol_wave0 = rbinom(n_accounts, size = n_fyp, prob = base_p),
    pol_wave1 = rbinom(n_accounts, size = n_fyp, prob = base_p + 0.08 *
    ↪  Z1),
    pol_wave2 = rbinom(n_accounts, size = n_fyp,
                       prob = base_p + 0.05 * Z1 + 0.08 * Z2)
  ) |>
  select(account, pol_wave0, pol_wave1, pol_wave2)

video_data <- expand.grid(
  video   = 1:25,
  hashtag = 1:3,
  round   = 1:2,
  account = 1:n_accounts
) |>
  left_join(select(assignments, account, Z1, Z2), by = "account") |>
  mutate(
    treated   = if_else(round == 1, Z1, Z2),
    # log-normal prevents negative numbers and creates "influencer" spikes
    followers = round(rlnorm(n(), meanlog = 8 + 0.5 * treated, sdlog =
    ↪  1.5)),
    # Poisson distribution for typical event counts
    likes     = rpois(n(), lambda = 250 + 100 * treated),
    comments  = rpois(n(), lambda = 15 + 10 * treated)
  ) |>
  select(account, round, hashtag, video, likes, comments, followers)
#
↪  ------------------------------------------------------------------------------

# merge assignments and outcomes into one analysis dataset
practicum_data <- assignments |>
  left_join(outcomes_data, by = "account")
```

```r
#-----------------------------------------
# 6. Covariate balance checks (F-statistic + RI p-value)
```

```r
# Check 2: is Z1 balanced on the baseline covariate pol_wave0?
declaration_Z1 <- declare_ra(N = n_accounts, m = n_treat)

balance_fun_z1 <- function(data) get_f(Z1 ~ pol_wave0, data)

ri_balance_z1 <- conduct_ri(
  test_function = balance_fun_z1,
  declaration   = declaration_Z1,
  assignment    = "Z1",
  sims          = n_sims,
  p             = "upper",
  data          = practicum_data
)
# print(summary(ri_balance_z1))

# Check 3: is Z2 balanced on pol_wave0 and pol_wave1?
balance_fun_2 <- function(data) get_f(Z2 ~ pol_wave0 + pol_wave1, data)

ri_balance_2 <- conduct_ri(
  test_function = balance_fun_2,
  declaration   = declaration_Z2,
  assignment    = "Z2",
  sims          = n_sims,
  p             = "upper",
  data          = practicum_data
)
# print(summary(ri_balance_2))
```

```r
#-----------------------------------------------
# 7. Descriptive results: mean (and SE) of the outcome by condition

tab_1 <- practicum_data |>
  group_by(Z2) |>
  summarise(
    n           = n(),
    outcome_mu  = mean(pol_wave2),
    outcome_se  = sd(pol_wave2) / sqrt(n()),
    wave0_mu    = mean(pol_wave0),
    wave1_mu    = mean(pol_wave1),
    .groups = "drop"
  )
# print(tab_1)

tab_2 <- practicum_data |>
```

```r
  group_by(randomize) |>
  summarise(
    n          = n(),
    outcome_mu = mean(pol_wave2),
    outcome_se = sd(pol_wave2) / sqrt(n()),
    .groups = "drop"
  )
# print(tab_2)

# Descriptive statistics for videos: likes, comments, followers
tab_3 <- video_data |>
  left_join(select(assignments, account, Z1, Z2), by = "account") |>
  mutate(condition = if_else(if_else(round == 1, Z1, Z2) == 1,
                             "Conservative-coded", "Apolitical (cockapoo)"))
                              ↪  |>
  group_by(round, condition) |>
  summarise(
    n_videos       = n(),
    mean_likes     = mean(likes),
    mean_comments  = mean(comments),
    mean_followers = mean(followers),
    # composite on the log scale per the PAP: log(x + 1) avoids -Inf at 0
    log_mean_likes     = log(mean_likes + 1),
    log_mean_comments  = log(mean_comments + 1),
    log_mean_followers = log(mean_followers + 1),
    .groups = "drop"
  )
# print(tab_3)
```

```r
#-------------------------------------------
# 8. Regression analysis
# H1: difference in means by assignment in randomization 2
# H2: outcome on number of rounds of conservative-coded content (0/1/2)
# Covariate adjustment uses only pre-treatment measures:
#   - Z2 models: pol_wave1 (PAP), then pol_wave0 + pol_wave1
#   - randomize model: pol_wave0 only (pol_wave1 is affected by Z1, part
 ↪  of randomize)

f_h1      <- pol_wave2 ~ Z2
f_h2      <- pol_wave2 ~ randomize
f_cov     <- pol_wave2 ~ Z2 + pol_wave1
f_full    <- pol_wave2 ~ Z2 + pol_wave0 + pol_wave1
f_h2_cov  <- pol_wave2 ~ randomize + pol_wave0
```

```r
m1 <- lm(f_h1,     data = practicum_data)
m2 <- lm(f_h2,     data = practicum_data)
m3 <- lm(f_cov,    data = practicum_data)
m4 <- lm(f_full,   data = practicum_data)
m5 <- lm(f_h2_cov, data = practicum_data)

# Generate table output (regular OLS standard errors; in the .Rmd, put
↪  this in a chunk with results='asis')
# stargazer::stargazer(m1, m2, m3, m4, m5, type = "latex", header = FALSE,
#                      no.space = TRUE, title = "Treatment Effect
↪  Estimates")
```

```r
#-----------------------------------------
# 9. Randomization inference

# --- H1: one-tailed (upper) test of the sharp null of no effect
ri_h1 <- conduct_ri(
  f_h1,
  declaration      = declaration_Z2,
  assignment       = "Z2",
  sharp_hypothesis = 0,
  sims             = n_sims,
  p                = "upper",
  data             = practicum_data
)
# summary(ri_h1)
# plot(ri_h1)

# 95% interval for the H1 estimate, assuming a constant effect equal to
↪  the ATE
ate_h1 <- ri_h1$sims_df$est_obs[1]
ri_h1_ci <- conduct_ri(
  f_h1,
  declaration      = declaration_Z2,
  assignment       = "Z2",
  sharp_hypothesis = ate_h1,
  sims             = n_sims,
  data             = practicum_data
)
ci_h1 <- quantile(ri_h1_ci$sims_df$est_sim, c(0.025, 0.975))

# --- H2: one-tailed (upper) test for the randomize variable.
randomize_est <- function(formula, data) coef(lm(formula, data =
↪  data))[["randomize"]]
```

```r
ri_randomize <- function(formula, data, p = c("upper", "two-tailed")) {
  p   <- match.arg(p)
  obs <- randomize_est(formula, data)
  sims <- replicate(n_sims, {
    data$randomize <- complete_ra(N = n_accounts, m = n_treat) +
      complete_ra(N = n_accounts, m = n_treat)
    randomize_est(formula, data)  # outcome fixed under the sharp null
  })
  pval <- if (p == "upper") mean(sims >= obs) else mean(abs(sims) >=
↪   abs(obs))
  list(estimate = obs, p_value = pval, sims = sims)
}

ri_h2  <- ri_randomize(f_h2, practicum_data, p = "upper")
est_h2 <- ri_h2$estimate
p_h2   <- ri_h2$p_value

# --- H2 adjusted for the baseline covariate pol_wave0: two-tailed
ri_h2_cov <- ri_randomize(f_h2_cov, practicum_data, p = "two-tailed")

# --- Covariate-adjusted (pol_wave1): two-tailed
ri_cov <- conduct_ri(
  f_cov,
  declaration      = declaration_Z2,
  assignment       = "Z2",
  sharp_hypothesis = 0,
  sims             = n_sims,
  p                = "two-tailed",
  data             = practicum_data
)

# --- Adjusted for pol_wave0 + pol_wave1: two-tailed
ri_full <- conduct_ri(
  f_full,
  declaration      = declaration_Z2,
  assignment       = "Z2",
  sharp_hypothesis = 0,
  sims             = n_sims,
  p                = "two-tailed",
  data             = practicum_data
)

# --- Results table: estimate, RI p-value, and the model F-statistic
results <- data.frame(
```

```
  model     = c("H1: difference in means (Z2)",
                "H2: randomize (Z1 + Z2)",
                "H1: Z2 + pol_wave1",
                "H1: Z2 + pol_wave0 + pol_wave1",
                "H2: randomize + pol_wave0"),
  test      = c("one-tailed", "one-tailed", "two-tailed", "two-tailed",
                "two-tailed"),
  estimate  = c(ate_h1, est_h2,
                ri_cov$sims_df$est_obs[1], ri_full$sims_df$est_obs[1],
                ri_h2_cov$estimate),
  ri_p      = c(summary(ri_h1)$upper_p_value,
                p_h2,
                summary(ri_cov)$two_tailed_p_value,
                summary(ri_full)$two_tailed_p_value,
                ri_h2_cov$p_value),
  f_stat    = c(get_f(f_h1,     practicum_data),
                get_f(f_h2,     practicum_data),
                get_f(f_cov,    practicum_data),
                get_f(f_full,   practicum_data),
                get_f(f_h2_cov, practicum_data))
)
# print(results)
# print(ci_h1)
```

```
#-----------------------------------------------
# 10. A graphical depiction of experimental findings in style of Coppock
↪  (2021)

coppock_plot <- function(data, group, labels) {
  data <- data |> mutate(group = factor(.data[[group]], labels = labels))
  # group means with regular OLS 95% confidence intervals
  fit <- lm(pol_wave2 ~ 0 + group, data = data)
  ci  <- confint(fit)
  summary_df <- data.frame(
    group     = factor(labels, levels = labels),
    pol_wave2 = unname(coef(fit)),
    conf.low  = unname(ci[, 1]),
    conf.high = unname(ci[, 2])
  )

  ggplot(summary_df, aes(group, pol_wave2)) +
    geom_point(
      data = data,
      position = position_jitter(width = 0.15, height = 0),
```

```
      alpha = 0.3
    ) +
    geom_point(size = 3) +
    geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width = 0) +
    theme_bw() +
    scale_y_continuous(limits = c(0, n_fyp), breaks = seq(0, n_fyp, by =
    ↪  10)) +
    theme(axis.title.x = element_blank()) +
    ylab("Explicitly political videos\n(of first 50 'For You' videos)")
}

plot_h1 <- coppock_plot(practicum_data, "Z2",
                        c("Apolitical (cockapoo)", "Conservative-coded"))
plot_h2 <- coppock_plot(practicum_data, "randomize",
                        c("0 rounds", "1 round", "2 rounds"))
# print(plot_h1)
# print(plot_h2)
```

References

Blair, Graeme, Jasper Cooper, Alexander Coppock, and Macartan Humphreys. 2019. "Declaring and Diagnosing Research Designs." *American Political Science Review* 113(3): 838–59. doi:10.1017/S0003055419000194.

Brown, Megan, James Bisbee, Angela Lai, Richard Bonneau, Jonathan Nagler, and Joshua Aaron Tucker. "Echo Chambers, Rabbit Holes, and Algorithmic Bias: How YouTube Recommends Content to Real Users." doi:10.2139/ssrn.4114905.

"Create and Manage Hide My Email Addresses in Settings on iPhone." https://support.apple.com/guide/iphone/create-and-manage-hide-my-email-addresses-iphcb02e76f7/ios.

Huszár, Ferenc, Sofia Ira Ktena, Conor O'Brien, Luca Belli, Andrew Schlaikjer, and Moritz Hardt. 2022. "Algorithmic Amplification of Politics on Twitter." *Proceedings of the National Academy of Sciences* 119(1): e2025334119. doi:10.1073/pnas.2025334119.

Liscomb, Megan. 2025. "This Woman Is Calling Out Ways the Far Right Targets Women and Girls Online." https://www.buzzfeed.com/meganeliscomb/alt-right-pipeline-for-women.

"TikTok Support." https://www.tiktok.com/support/faq_detail?id=7581821549574052364.